

Lab 4: Stringing Together Physics and Physiology

Physical Models: Speed of Sound, Weber-Fechner Law

Analysis Tools: Random and Systematic Uncertainty, t-score, propagation of uncertainty

Experimental Systems: Ukulele

Equipment: Audacity, Phyphox, meterstick, tape measure, clip on tuner, microphone, caliper, micrometer, hook scale, capos

Safety Concerns: Remember that since loudness has a logarithmic relationship with intensity, incremental changes in loudness correspond to exponentially increasing changes in intensity. A sound can go from being safe to dangerous very quickly! And long exposure times to lower intensities can still result in hearing losses. If you're interested in having your hearing checked (for free!) you can sign up using the QR code to the right. **You'll even get 3 points of extra credit on this lab!**

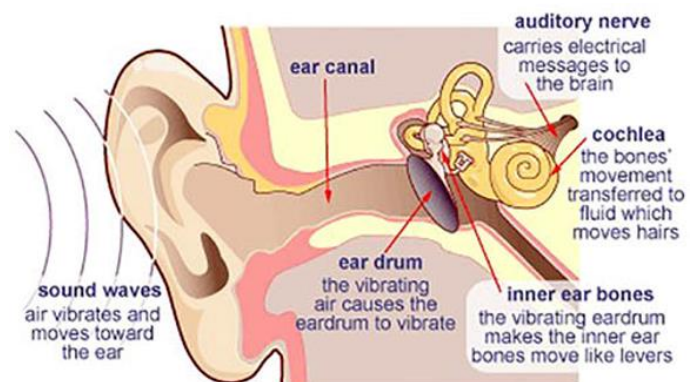
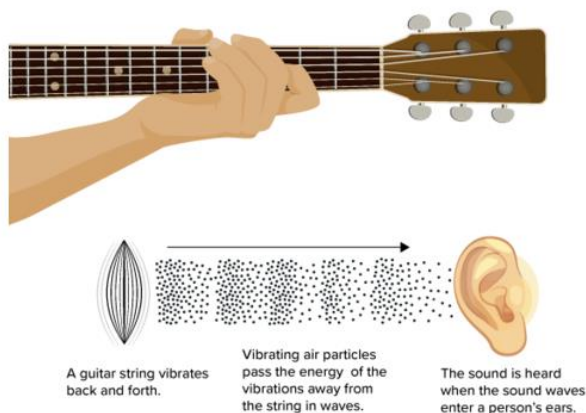


Introduction

This week is a special lab which is coordinated with the Department of Speech, Language, and Hearing Sciences and the Moody College of Communication Speech and Hearing Center.



We'll be investigating the *physics of how sound is produced and travels* but also the *physiology of how sound is perceived*. You'll even get to try out some simple hearing self-tests! Audiology, hearing science, acoustics, and many other fields are integrated with physics and encompass a variety of important applications in healthcare and music.



Extra Credit Opportunity: Under “Safety Concerns” and on canvas you’ll find a signup sheet for a free hearing exam. If you sign up you’ll earn 3 points of extra credit on this lab!

Part 1. The Speed(s) of Sound

As discussed in the prelab, sound is a mechanical vibration which travels at a speed that depends on the properties of a medium. For example, a vibrating string has a speed of sound given by:

$$c_{s,string} = \sqrt{T/\mu}$$

Where T is the tension force and μ is the mass per length of string. Meanwhile for air the speed of sound is about:

$$c_{s,air} = \sqrt{\gamma P/\rho} \approx 344 \text{ m/s}$$

As a vibration moves from one medium to another, the velocity will change. However, the frequency (which we perceive as pitch) remains constant. These quantities are related by the formula:

$$f\lambda = c_s$$

Quick Check: Try plucking any two of the strings on the Ukulele. Which string has the higher frequency/pitch? Which has the lower?

When you pluck a string, the vibration travels from the string, through the wooden bridge, to the resonator chamber, which vibrates the surrounding air, which then travels into your ear. The frequency remains constant even as the velocity changes.

Do you think the two strings you plucked have the same speed of sound or not? Make a prediction for how the speeds of the two strings relate to each other. Put this in your methods section.

The Model and its Assumptions: If we combine the previous equations with the wavelengths of the vibrating string, we get the following relation:

$$2fL = c_{s,string} = \sqrt{T/\mu} = \text{constant}$$

Where f is the peak (loudest) frequency and L is the length of the string.

This means that if two different strings have different sound velocities, at least one of the following must be true:

1. The tension of the strings are different.

2. The mass per length are different because the strings are made of different materials
3. The mass per length are different because the strings have different diameters
4. The model is wrong or imprecise (due to extra material forces, changes in length during vibration, or friction) or the string was plucked in a way such that the fundamental frequency f was not the peak frequency

Designing Your Experiment: Your goal is to design a high precision measurement of the speed of sound of the two strings you plucked in the “Quick Check”.

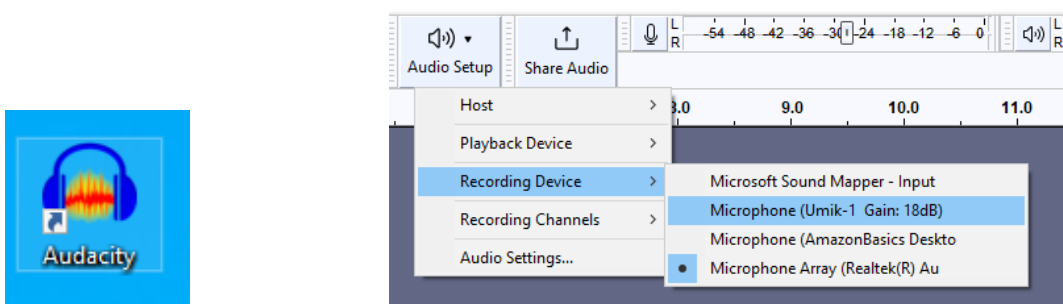
Since we have not tuned the Ukulele, any outcome is possible!

If you find the velocities are different, try to assess which of the assumptions 1-4 are violated. If you find they are the same, make an assessment of if 1-4 are reasonable in this case. A simple quick check is sufficient (e.g. check if the tensions or widths are the same or different).

If you use propagation of uncertainty, here are the needed derivatives:

$$\frac{\partial c_s}{\partial L} = 2f \frac{\partial c_s}{\partial f} = 2L$$

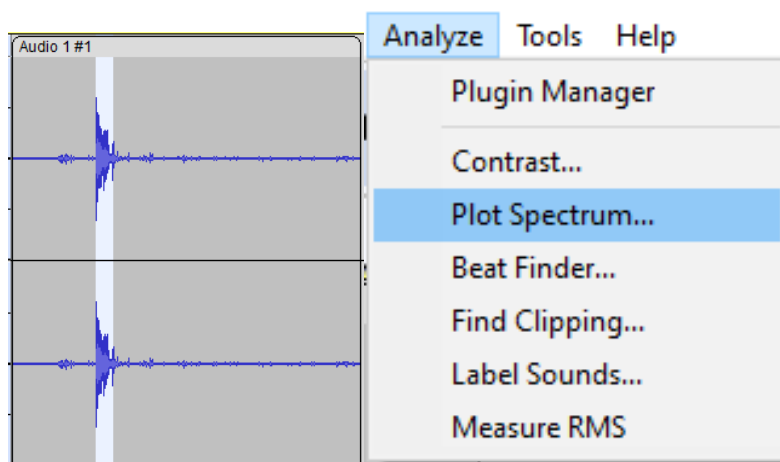
To measure frequency, you can use either your cell phone app (e.g. PhyPhox) or the microphone and the Audacity program. To open Audacity, click the icon. Make sure to select the UMIK audio input.



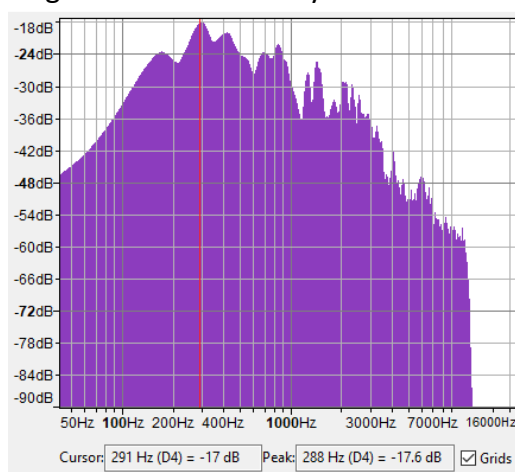
You can select “record” to record a sound:



You can highlight a sound you’d like to analyze. Highlight it and select “Plot Spectrum”



You'll see a full spectrum. By moving the cursor around you can find the frequencies of the peaks:



Our assumption is that the peak (loudest) frequency is the one we want.

Check in with Another Group: Briefly discuss your plans with another group. They may provide you with additional ideas and feedback. Note one similarity and one difference between your approaches.

Conducting Your Experiment: Conduct your experiment as planned.

Forming a Conclusion: How do the velocities of the strings compare? What evidence do you have (about the tension, mass density, etc.) that could explain this?

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

Part 2. The Physiology of Sound Perception

Hearing Self-Check: Take a tuning fork and gently hit it against the table until it starts to ring. Then, put the bottom of the stem against the center of your forehead. Repeat this with a tuning fork at a different resonant frequency. Does the higher frequency tuning fork sound as loud as the lower frequency tuning fork?

When you place the vibrating tuning fork to your forehead, part of the sound that you hear comes from the vibrations (*traveling through the bone of your skull!*) to your cochlea. This is called **bone conduction**. As you observed in part one, low frequency waves have much longer wavelengths than high frequency waves, meaning that low frequency sounds can travel the same distance in fewer cycles. Each cycle, sound intensity is lost to dissipative effects, and so sounds **attenuates** as it travels. Roughly speaking, fewer cycles means less attenuation.

Given that, when you speak, there is a range of frequencies produced by your vocal folds, how do you think this phenomenon relates to why your voice sounds different in your head than in a recording? Is your voice as you hear it higher or lower? Answer in your lab notes.

Quick check: As you learned in the prelab, perception is distinct from physical stimulus. For frequency perception, frequency is the physical stimulus while pitch is the perceptual counterpart. While they are related, there is a complex relationship between the two. One of the most common examples of perceptual units for pitch is the musical notes within the chromatic scale. Each note is perceived to increase by an equal amount between notes (ex: C4 to C4# is the same increase as C4# to D).

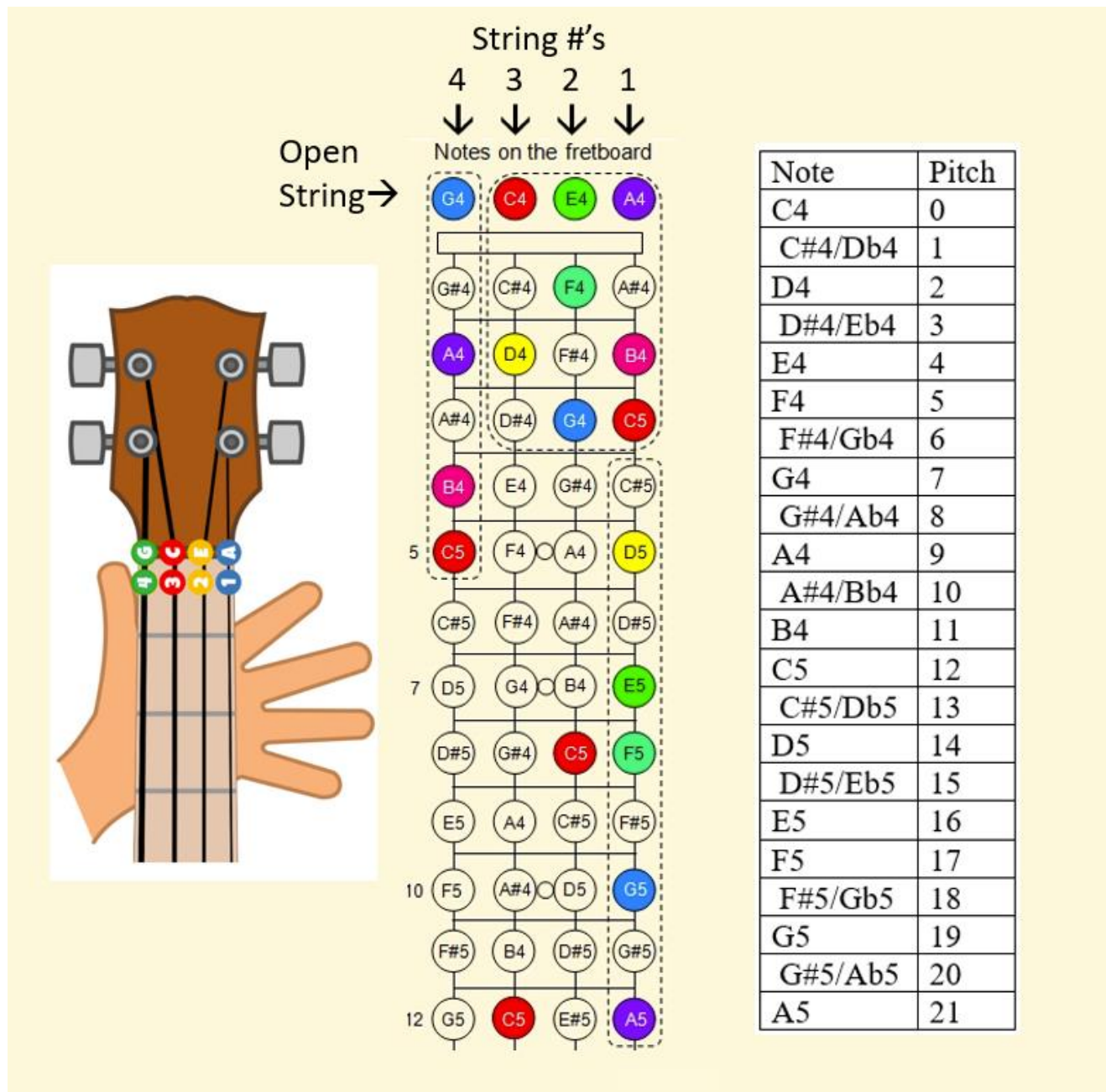
Use the tuner to tune your Ukulele the best that you can so that the strings from top to bottom (when you are holding it) are G4, C4, E4, and A4.



Go ahead and play some notes at equal intervals. This can be playing the ukulele fret by fret (G4 to G4# to A4#) or octave by octave (A4 to A5 or G4 to G5 etc.). Use the picture of the ukulele fretboard below to

identify these intervals. When you play these intervals, do they sound equally spaced in pitch to you? If so, we can treat musical intervals as corresponding to our pitch perception.

Defining Variables: We will measure pitches by counting notes up from the lowest note on the Ukulele, which is C4, the open 3rd string. We'll call this $P = 0$. It has a frequency $f_{C4} \approx 261 \text{ Hz}$. See the note map of the fretboard below and the table of pitches to read off which note is which pitch:



For example, if you hold down the 5th position on the 4th string, the note is C5. In the table this pitch is $P = 12$.

The Two Models: In comparing pitch and frequency, we will consider the following three possibilities, using the variables defined above:

1. The relationship between pitch and frequency is *linear*, that is:

$$P = k_L(f - f_{c4})$$

2. The relationship between pitch and frequency is *logarithmic* and obeys the *Weber-Fechner Law*:

$$P = k_{WF} \log \frac{f}{f_{c4}}$$

3. The relationship is more complicated than either of these.

Where again we are treating $P=0$ as $C4$ and $f_{c4} \approx 261 \text{ Hz}$.

Designing your experiment: The goal here is to design an experiment that investigates how frequency and pitch are related. It's up to you how to do this. One suggestion is to check if either, both, or neither of:

$$k_L = \frac{P}{f - f_0} \quad k_{WF} = \frac{P}{\log\left(\frac{f}{f_0}\right)}$$

are constant.

You'll need to decide how to assign an uncertainty to pitch, as well as to frequency.

The needed uncertainty propagations are:

$$\delta k_L = \sqrt{\left(-\frac{P}{(f - f_0)^2} \delta f\right)^2 + \left(\frac{1}{f - f_0} \delta P\right)^2}$$

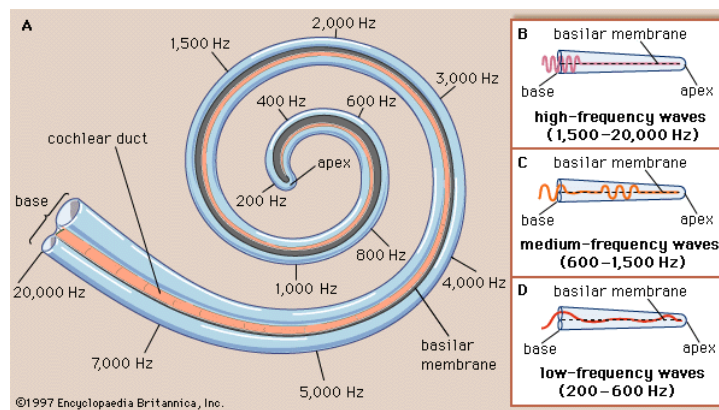
$$\delta k_{WF} = \sqrt{\left(-\frac{P}{f \log(f/f_0)^2} \delta f\right)^2 + \left(\frac{1}{\log(f/f_0)} \delta P\right)^2}$$

Forming a Conclusion: Were both models, one model, or neither model consistent with your data? What does this mean about pitch perception? How does it compare with the perception of, for example, loudness?

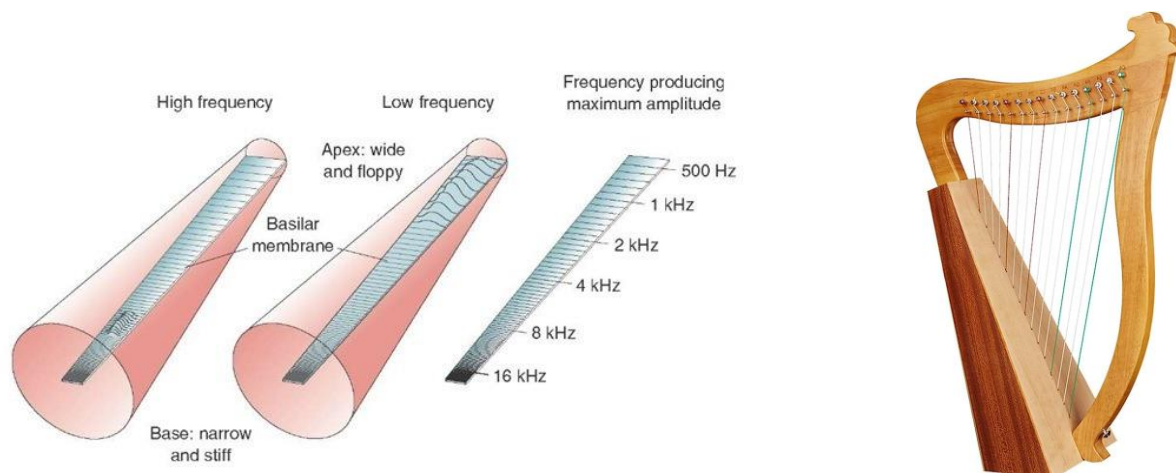
Synthesis

The **basilar membrane** is the membrane within the cochlea that separates complex sound into its individual frequencies. The images below show how the basilar membrane is **tonotopically organized**,

which just means that each place along the membrane has a different resonant frequency, and these resonant frequencies decrease from the base to the apex.



The picture above shows the basilar membrane within the cochlea, where it has a spiral shape



The picture on the left shows the basilar membrane as if it were removed from the cochlea and unrolled. The picture on the right shows a harp with strings of varying lengths and masses.

A Final Question: You saw in the first part of the lab that mass per length of the string and tension of the string can influence the sound velocity and frequency. If we think of the basilar membrane as many strings bunched together, how do these two factors explain why the apex of the basilar membrane responds to low frequencies and the base responds to high frequencies?

We know that the perception of frequencies follows a power law. You determined the nature of this law in Part 2 of the lab. For example, if pitch obeys the Weber-Fechner law, then an incremental increase in pitch perception requires an exponential increase in frequency. Look at the tonotopic organization of the basilar membrane in the first picture. How does the non-linear relationship between frequency and pitch perception relate to how the resonant frequencies change with equidistant spacing along the membrane? Would the resonant frequencies change proportionally with distance, or in another way? How?